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Dynamic pricing and reference price effects

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ABSTRACT

Dynamic pricing enables online retailers to dynamically adjust prices in response to demand and supply conditions. However, dynamically changing prices may also affect how consumers assess current retail prices compared to their internal reference prices. This paper develops propositions for the influence of dynamic pricing on the internal reference price effect based on a reduced demand model, and tests these propositions with a field experiment and an online lab experiment. Both experiments systematically manipulate two dimensions of dynamic price variation (amplitude and frequency). We find that customers react strongly to price increases but not to price reductions. However, higher amplitude, as well as the higher frequency of price changes, attenuate the internal reference price effects in the field experiment. The online lab experiment provides additional insights in a more controlled setting measuring the internal reference price and additional covariates. Therefore, dynamic pricing can provide an additional benefit for retailers by attenuating potentially detrimental reference price effects on demand.

1. Introduction

For decades, the practice of dynamic pricing has been applied by airlines and hotels to balance demand and supply as well as avoid unused capacity of perishable products (Gallego & van Ryzin, 1994). Recently, dynamic pricing has been extended to durable products in online retailing. This practice is facilitated by the increased availability of customer and competitor data as well as the use of algorithmic price setting (Elmaghraby & Keskinocak, 2003; Fisher, Gallino, & Li, 2018).

The application of dynamic pricing is promoted by increased competition and high price transparency on digital marketplaces such as Amazon. Consumers can choose between different sellers offering the same product, and digital marketplaces facilitate such price comparison (Chen, Mislove, & Wilson, 2016). Additionally, consumers can compare prices on price-comparison websites and observe historic price developments on price-tracking websites (Drechsler & Natter, 2011).

Nevertheless, dynamic pricing can lead to consumers' backlashes due to comparisons of transactions (Feinberg, Krishna, & Zhang, 2002; Haws & Bearden, 2006; Xia, Monroe, & Cox, 2004). Therefore, the question arises, how to implement dynamic pricing successfully. For example, Weisstein, Monroe, & Kukar-Kinney (2013) show that using various price-framing tactics can reduce consumers' perceived similarity of transactions and thereby improve consumer responses.

Consumers compare recent retail prices with memorized past prices (internal reference prices) and the prices of alternatives at the point of purchase (external reference prices) (Kalyanaram & Winer, 1995). Dynamic pricing may affect how consumers assess this comparison of current prices with their reference prices (Gönsch, Klein, Neugebauer, & Steinhardt, 2013). While research on reference price effects has been extensive, there has not been much focus on the impact of dynamic pricing on the reference price effect (see Elmaghraby & Keskinocak (2003) for an overview).

Dynamic pricing may strengthen or attenuate internal reference price effects. On the one hand, dynamic pricing can increase attention to prices (Vanhuele & Drèze, 2002) and thus strengthen internal reference price effects. On the other hand, dynamic pricing can distort consumer price evaluation because of increased difficulty or uncertainty (Haynes, 2009; Mazumdar & Jun 1992): The large number of price points may hinder conscious deliberation of alternatives (Dijksterhuis, Bos, Nordgren, & van Baaren, 2006). Thus, internal reference prices could play a reduced role in consumption decisions with dynamic prices.

The magnitude of internal reference price effects is an important consideration for retailer strategy. From a retailer's perspective, it can be advantageous to attenuate internal reference price effects when prices rather increase, e.g., in the case of firms using low introductory prices that may be later increased. On the contrary, it would be

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beneficial to strengthen internal reference price effects when prices usually decrease, e.g., in the case of price skimming strategies (Spann, Fischer, & Tellis, 2015).

Our study contributes to the literature by empirically analyzing the effect of dynamic pricing on internal reference price effects. We develop three propositions for the influence of dynamic pricing on the internal reference price effect derived from a reduced demand model. We test our propositions in a field experiment at a durable products online retailer and an online lab experiment.

This paper conceptualizes the price variation of dynamic prices according to two key dimensions. First, the *amplitude* of price changes captures the magnitude of price changes. Second, the *frequency* of price changes captures how often prices change. The field experiment includes 181 durable electronic products. We randomly assign products to four different price variation treatments (based on amplitude and frequency) and estimate internal reference prices. The online lab experiment captures the key aspects of the field experiment by randomly assigning participants to different price variations (based on amplitude and frequency). In addition, the online lab experiment measures participants' internal reference prices as well as covariates such as perception of trust, quality, and opportunism towards the retailer.

We use the estimated (measured) internal reference prices of the field (online lab) experiment to analyze the impact of dynamic pricing on internal reference price effects. Our results show an asymmetric internal reference price effect for the demand for durable products. Losses, i.e., a current price higher than the predicted or observed internal reference price, have a substantial negative effect on purchase behavior. However, we find no such effect for gains, i.e., a current price lower than the internal reference price. More importantly, the field experiment finds that higher amplitude and / or higher frequency of prices attenuates internal reference price effects in the loss domain. The online lab experiment provides additional support for the attenuating effect of dynamic pricing.

2. Related literature

In this section, we summarize literature on dynamic pricing, as well as its impact regarding the amplitude and frequency of price changes, on the relationship between reference prices and demand.

2.1. Dynamic pricing

Research on dynamic pricing focusses on different directions (see also the summary by Neubert (2022)). While several studies use formal models to analyze the consequences of dynamic pricing (Huang, Luo, & Xia, 2019; Jing, 2011; Levin, McGill, & Nediak, 2009), we focus on the recent empirical literature on dynamic pricing (see Table 1) in comparison to this paper.

An important stream of this research tries to detect and characterize dynamic pricing. For example, Chen et al. (2016) detect and analyze more than 500 dynamic pricing sellers on the Amazon marketplace, Spann et al. (2015) characterize dynamic pricing strategies in the market of consumer electronics, and Hillen & Fedoseeva (2021) analyze dynamic pricing activities for online grocery stores.

Since experimental price variations are associated with profit consequences in practice, most empirical studies focus on observational data, e.g., Schmidt, Bornschein, & Maier (2020) and Wamsler, Natter, & Algesheimer (2022), or laboratory experiments, e.g., Abrate, Nicolau, & Viglia (2019) and Hillen & Fedoseeva (2021). The study of Fisher et al. (2018) is one of the few examples which randomly varies the prices of products in a field experiment. They develop a best-response pricing strategy based on a consumer choice model and incorporate competitors' actions, and supply parameters such as procurement costs, margin target, and manufacturer price restrictions.

Another research stream studies the consequences of dynamic pricing on price fairness perceptions and how retailers can mitigate these negative effects (Alderighi, Nava, Calabrese, Christille, & Salvemini, 2022; Schmidt et al., 2020; Wamsler et al., 2022). Abrate et al. (2019) directly estimate the impact of price variation on the performance of the retailer (i.e., in their case the revenue of available hotel rooms). They show that higher variability in prices is associated with higher revenues. Similarly, Zhang et al. (2020) show that dynamic pricing in the form of discounts can increase demand in the short run. However, this creates incentives for strategic consumer behavior such as adding more products to the shopping cart to elicit more shopping cart promotions.

Our study contributes to the understanding of how dynamic pricing can affect customer behavior related to internal reference price effects. We focus on customer demand and study how dynamic pricing attenuates internal reference price effects with experimental data from the laboratory and the field.

We expect that higher price variation in dynamic pricing may mitigate unfavorable consumer reactions to price changes. This prediction would offer a behavioral explanation for the contraction between Alderighi et al. (2022) and Haws & Bearden (2006)'s findings that while dynamic pricing is perceived as unfair, it is extensively applied in practice (Chen et al., 2016) and can yield higher revenues (Abrate et al., 2019).

2.2. Asymmetric internal reference prices

Previous research has shown that consumer reactions to a price not only depend on the actual price but are also influenced by the comparison of this price to a reference point (Kalyanaram & Winer, 1995). In line with the theory of reference-dependent preferences, the reference price operates as a reference point to which consumers experience gains or losses. Gains occur when the current price is lower than the reference point, for example, when prices decrease. Losses develop when the current price is higher than the reference price, for example, when prices increase (Kalyanaram & Winer, 1995).

Empirical studies analyzing whether gains or losses weigh more strongly in consumer decisions primarily investigate scanner panel data of consumable products and report mixed results. Greenleaf (1995) finds a stronger effect of gains than losses on peanut butter sales. Krishnamurthi, Mazumdar, & Raj (1992) differentiate consumers between brand switchers and brand loyals for frequently bought products. For brand switchers, the impact of a gain is stronger than that of a loss. Demand changes for brand loyals seem to be equal for losses and gains.

Prospect theory provides a theoretical explanation for the phenomenon of consumers being more sensitive to losses than to gains (Kahnemann & Tversky, 1979; Thaler, 1985). Several empirical studies found a stronger effect of losses than gains for differing products. For example for ground coffee (Kalyanaram & Little, 1989), sweetened and unsweetened drinks (Kalyanaram & Little, 1994), yogurt (Mayhew & Winer, 1992), and orange juice (Hardie, Johnson, & Fader, 1993), losses seem to loom larger than gains. Putler (1992) finds a significant negative effect for losses but an insignificant positive effect for gains in the category of eggs. Bell & Bucklin (1999) investigate the impact of reference-dependent preferences by studying purchase postponements and purchase accelerations. They find that the positive effect of a price decrease cannot compensate for the negative effect of a price increase for saltine crackers and liquid laundry detergent.

There are fewer empirical results on internal reference price effects in the case of durable products. However, the characteristics of durable products differ from characteristics of non-durable products (Elmaghraby & Keskinocak, 2003). For example, consumer electronics are characterized by frequent new product introductions and rather short product life cycles (Kelly, Papanikolaou, Seru, & Taddy, 2021; Spann et al., 2015). Further, durable products tend to be associated with a higher purchase risk due to higher prices, a higher functional complexity (Roberts & Urban, 1988), and higher involvement (Laurent & Kapferer, 1985). Consumers can account for this risk by more intensive search (Moorthy, Ratchford, & Talukdar, 1997) and higher information

Table 1
Recent empirical research on dynamic posted pricing.

Study	Theoretical Framework	Data Source			Dynamic Pricing Approach	Accounts for ERPs	Accounts for IRPs
		Observational	Online lab experiment	Field experiment			
Abrate et al. (2019)	Hedonic revenue model with price level, price variability and quality attributes as explanatory variables	✓			Dynamic pricing (=price variability) increases revenue		
Alderighi et al. (2022)	Price fairness perceptions based on customer evaluation of their stay at a hotel	✓			Dynamic pricing (=price variability) decreases fairness perceptions		✓
Chen et al. (2016)	Methodology for detecting algorithmic pricing, and applying it to observational data to analyze dynamic pricing prevalence and behavior on Amazon Marketplace	✓			Describe characteristics of sellers, which use dynamic pricing (=prices respond quickly to competitors' prices)		
Fisher et al. (2018)	Consumer-choice model that incorporates all of the competing options consumers have	✓		✓	Dynamic pricing (=prices respond quickly to competitors' prices) increases revenue	✓	
Guizzardi, Mariani, & Stacchini (2022)	Temporal distance according to construal theory conceptualized in days of advance booking of hotel rooms. Model reflects the dynamic pricing behavior of supply and the seasonal oscillation pattern of demand	✓	✓		Dynamic pricing (=price variability) influences consumers' quality perceptions		✓
Hillen & Fedoseeva (2021)	Price rigidity of grocery products changes with increasing online retailing	✓			Dynamic pricing (=small and frequent price adjustments) is introduced in the online grocery market at Amazon Fresh		
Huang et al. (2019)	Dynamic pricing model which incorporates ex-ante information about the demand of products based on initial assessments and sellers' learning	✓			Dynamic pricing (=price updates due to demand learning) increases revenue		
Schmidt et al. (2020)	Attribution theory of price changes in dynamic and personalized pricing is influenced by whether consumers approve data collection		✓		Consumers' consent with tracking of their browsing behavior leads to stronger price change attribution to themselves which avoids detrimental decreases in fairness perceptions for dynamic pricing (=frequent price changes)		✓
Spann et al. (2015)	Classification of dynamic pricing strategies in skimming, penetration, and market pricing corresponding to product attributes, market trends, and product age	✓			Dynamic pricing in form of different strategies (such as skimming, penetration, and market pricing) is applied under corresponding market, firm, and brand characteristics		
Wamsler et al. (2022)	Dynamic pricing influences behavioral consequences due to perceived distributive fairness. Pricing authority moderates this effect		✓		Dynamic pricing (=price differences due to demand) affects distributive fairness perceptions, repurchase intentions, tipping decision, and tipping amount		
Zhang et al. (2020)	Short and long-term effects of dynamic pricing implemented via discounts focusing on customer engagement and their strategic behavior			✓	Dynamic pricing (=discounts on products in the shopping cart) increases customer engagement, sales, and strategic customer behavior		✓
This paper	Demand depends on reference price effects affected by dynamic pricing		✓	✓	Dynamic pricing (=price change frequency and/or amplitude) attenuates (negative) internal reference price effects	✓	✓

processing (Beatty & Smith, 1987). Therefore and due to less frequent purchases, we presume durable products' internal reference prices more likely stem from product search than from previous purchases. Building on previous literature and empirical generalizations (Kalyanaram & Winer, 1995), we expect stronger internal reference price effects for losses than for gains also in the case of durable products.

2.3. Impact of amplitude and frequency on internal reference price effects

Uncertainty can increase the acceptable price range of consumers (Dickson & Sawyer, 1990; Winer, 1986). In an experimental study, Mazumdar & Jun (1992) manipulated uncertainty by presenting either fluctuating or stable prices. This induced price uncertainty increased the gap between participants' tentatively allocated budget and their maximum price. Uncertain consumers seemed to have a higher tolerance to prices exceeding their budget (i.e., a loss) than certain consumers. Accordingly, Caputo, Lusk, & Nayga (2020) argue that infrequent purchases and high price variability of a product cause reference price uncertainty. They show that this higher reference price uncertainty reduces the impact of this reference price by smoothening the kink in the demand curve. Similarly, Bodur & Arora (2014) show that low confidence in one's reference price reduces the impact of this reference price. Further, they show that this effect is stronger in the loss domain.

Hence, we expect that high amplitude in price changes increases the acceptable range of prices directly and indirectly via reference price uncertainty since amplitude is a main component in price variability. In

line with Caputo et al. (2020), we conclude that high amplitude in price changes attenuates the internal reference price effect in the decision process of customers.

Since the frequency of price changes is another component of price variability, high frequency can also increase uncertainty. Additionally, high frequency in price changes increases information regarding a product as well as the number of potential decision alternatives: A consumer may decide to buy now, later, or not at all for any price scenario. Consequently, dynamic prices increase the number of alternatives in a choice set. Therefore, when prices change more frequently, the alternatives in a choice set deviate rapidly, and consumers have less time to process the information of a choice set. Chernev, Böckenholt, & Goodman (2015) conclude from a meta-study that time constraint mediates how assortment size influences choice overload. Hence, consumers evaluating an increased number of decision alternatives in less time may suffer from choice overload. Relatedly, Bettman, Luce, & Payne (1998) show that a limited evaluation period forces consumers to engage in a less systematic assessment of the accessible alternatives.

Hence, we expect that frequent price changes reduce the systematic assessment of prices relative to the internal reference price due to increased choice overload (Chernev et al., 2015). Additionally, high price frequency, similar to high price amplitude, increases the price variability and therefore the reference price uncertainty (Caputo et al., 2020). We conclude that this results in an attenuated impact of internal reference prices in the decision process of consumers.

3. Model and propositions

We develop a formal demand model that accounts for the theoretical insights derived from the related literature discussed in Chapter 2. Following previous research (Kalwani, Yim, Rinne, & Sugita, 1990), we account for an asymmetric impact of reference price effects in $Gain_{jt}$ and $Loss_{jt}$ (regarding product j at time t):

$$Gain_{jt} = \begin{cases} Ref.Price_{jt} - Price_{jt} & if\ Ref.Price_{jt} \geq Price_{jt} \\ 0 & otherwise \end{cases} \tag{1}$$

$$Loss_{jt} = \begin{cases} Price_{jt} - Ref.Price_{jt} & if\ Ref.Price_{jt} < Price_{jt} \\ 0 & otherwise \end{cases} \tag{2}$$

To account for greater effects of losses than same-sized gains, we include a parameter λ for loss aversion with $\lambda \geq 1$. Further, to incorporate the impact of dynamic pricing on the reference price effect we include the amplitude σ_j^a and the frequency σ_j^f of price changes of product j. Both σ_j^a and σ_j^f are defined on a range from low amplitude/frequency to high amplitude/frequency:

$$Amplitude\ \sigma_j^a \in [0, 1] \tag{3}$$

$$Frequency\ \sigma_j^f \in [0, 1] \tag{4}$$

Our reduced demand model (5) incorporates a baseline demand a_j, as well as the loss aversion parameter λ to account for an asymmetric internal reference price effect on demand. The model assumes in line with previous research that amplitude and frequency increase information, uncertainty in prices, and the acceptable range of prices. Therefore, increasing amplitude and frequency attenuate the impact of perceived gains and perceived losses.

$$D_{jt} = \begin{cases} a_j - p_{jt} + \left(1 - \sigma_j^a\right)*\left(1 - \sigma_j^f\right)*\left(Ref.Price_{jt} - Price_{jt}\right) & if\ Ref.Price_{jt} \geq Price_{jt} \\ a_j - p_{jt} - \lambda*\left(1 - \sigma_j^a\right)*\left(1 - \sigma_j^f\right)*\left(Price_{jt} - Ref.Price_{jt}\right) & if\ Ref.Price_{jt} < Price_{jt} \end{cases} \tag{5}$$

We derive three empirically testable propositions from the model above. In the following, we present the propositions that we test in the experimental studies outlined below.

Proposition 1. *Demand increases in the internal reference price.*

$$\frac{\partial D}{\partial Ref.Price} = \begin{cases} (1 - \sigma_a)\left(1 - \sigma_j^f\right) \geq 0\ if\ Ref.Price_{jt} \geq Price_{jt} \\ \lambda(1 - \sigma_a)\left(1 - \sigma_j^f\right) \geq 0\ if\ Ref.Price_{jt} < Price_{jt} \end{cases} \tag{6}$$

This proposition follows from the notion that an increase of the internal reference price results in either an increase in gains or a decrease in losses (see equations (1) and (2)).

Proposition 2. *Higher amplitude in price changes attenuates the impact of internal reference price effects on demand.*

$$\frac{\partial\partial D}{\partial Ref.Price\partial\sigma_a} = \begin{cases} -\left(1 - \sigma_f\right) \leq 0\ if\ Ref.Price_{jt} \geq Price_{jt} \\ -\lambda*\left(1 - \sigma_f\right) \leq 0\ if\ Ref.Price_{jt} < Price_{jt} \end{cases} \tag{7}$$

Proposition 3. *Higher frequency in price changes attenuates the impact of internal reference price effects on demand.*

$$\frac{\partial\partial D}{\partial Ref.Price\partial\sigma_f} = \begin{cases} -(1 - \sigma_a) \leq 0\ if\ Ref.Price_{jt} \geq Price_{jt} \\ -\lambda*(1 - \sigma_a) \leq 0\ if\ Ref.Price_{jt} < Price_{jt} \end{cases} \tag{8}$$

Next, we test the three propositions derived from our model in two experiments: a field experiment where we observe sales data in reaction to manipulated prices, as well as an online lab experiment, which allows measuring participants' internal reference prices and additional control variables.

4. Study 1: Field experiment

To test the model's predictions, we conducted a field experiment with a European durable products retailer. We chose an electronics retailer since products in this category are durable and assumed to entail sufficient consumer deliberation (e.g., price search and comparison) before buying. In addition, dynamic pricing has become a common practice for durable products in online retailing (Mantin & Rubin, 2016).

4.1. Experimental design and data

We introduce a 2 (amplitude) × 2 (frequency) full factorial between-products design (Fig. 1). The first experimental factor (amplitude) differentiates between high (~8%) and low (~5%) amplitude in price changes. The second experimental factor (frequency) differentiates between high (every ~ 2nd day) and low (every ~ 4th day) frequency of price changes.

All price changes are relative to their product-specific baseline price, which is the median price in the month prior to the experiment. The specific experimental factors were determined in discussion with the

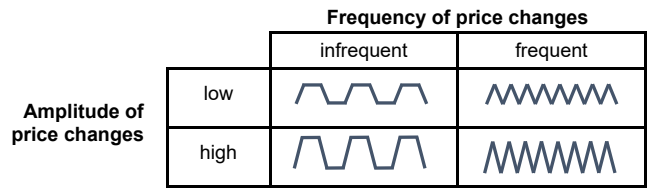

Fig. 1. Illustration of 2 (amplitude) × 2 (frequency) full factorial experimental design.

project partner on the basis of amplitude and frequency of previous prices.[1]

The 181 durable products were randomly assigned to the resulting four treatment groups. The products are from several distinct product categories within consumer electronics. The retailer classifies its products into four different groups and thereby ranks their popularity. The products in this experiment are all from the two intermediate popularity groups to avoid outliers in either direction. We observe prices, competitor prices, delivery time, and product purchases for 47 days, including the experimental period of 17 days.

We tested the integrity of our randomization procedure by using a probit panel model with random effects. The non-experimental control variables and the absolute price were used to explain the products' allocation to each of the two treatment variables (i.e., high versus low amplitude and high versus low frequency). We found that none of these variables explains this allocation, thus suggesting a successful randomization procedure (see Table A1 in Appendix A for details).

4.2. Estimation strategy

Our primary goal is to investigate the impact of dynamic prices on the effect of internal reference prices on demand. Since we do not observe the internal reference prices of consumers in the field experiment, the first key component in our estimation strategy is the estimation of internal reference prices.

Kalwani et al. (1990) estimate expected prices, which they define as equivalent to internal reference prices. They use the six last observed prices of coffee, the overall price trend in their observations (i.e., coffee brands for 65 weeks), and study-specific additional variables. Natter & Hruschka (1998) simplified this approach with only one lagged price and a trend variable. We adopt the approach of Kalwani et al. (1990) to estimate an internal reference price for each product j and day t with historic prices as well as category-specific price trends in observed prices. We capture the historic prices of the focal product with the variable $Past.Prices_{jt}$, which is the weighted moving average of prices during the last six days for each product and day. More recent prices receive a higher weight than more distant past prices:

$$Past.Prices_{jt} = \frac{\sum_{m=1}^{6} w^{m-1} * Price_{j(t-m)}}{\sum_{m=1}^{6} w^{m-1}} \tag{9}$$

The parameter w^{m-1} represents the weight applied to the mth lagged price observation of product j. As w decreases, consumers weigh recent prices more strongly. There is no theoretical basis for choosing the value of w. Following Kalwani et al. (1990) and Greenleaf (1995), we define w by searching for the parameter value that maximizes the explanatory power of our reference price model in equation (10). In this procedure, for each product and day, $Past.Prices_{jt}$ were calculated using equation (9) while varying w in increments of 0.1. Equation (10) yields the best fit for $w = 0.4$ (see Table A2 in Appendix A for estimations with varying weights). The second factor of the internal reference price prediction is category-specific $Price.Trend_j$. Consequently, we specify the following internal reference price equation:

$$\begin{gathered} Ref.Price_{jt} = \alpha_0 + \alpha_1 Past.Prices_{jt} + \alpha_2 Price.Trend_j + u_j + \varepsilon_{jt} \\ \boldsymbol{\varepsilon} \sim \mathrm{N}\left(0, \sigma_\varepsilon^2\right),\ \boldsymbol{u} \sim \mathrm{N}\left(0, \sigma_u^2\right) \end{gathered} \tag{10}$$

Here, u_j represents the unobserved product-specific random effects, which are supposed to be uncorrelated across products and correlated within products over time. For each day t and product j, the comparison of the predicted internal reference price and the current price results in either a positive or a negative difference. Previous research has commonly modeled this difference as a gain or loss (Kalwani et al., 1990). We follow this approach and calculate the variables $Gain_{jt}$ and $Loss_{jt}$ as in equation (1) and (2).

The second key component of our model is the moderating role of dynamic pricing on internal reference price effects. We capture the intensity of dynamic pricing based on the price variability according to two distinct dimensions: first, how often a price changes and, second, the strength of a price change. Therefore, we decompose the variation of price into the amplitude and frequency of price changes operationalized by the variables $High.Amplitude_{jt}$ and $High.Frequency_{jt}$. The interactions with $Gain_{jt}$ and $Loss_{jt}$ capture their moderating impact on internal reference price effects.

Furthermore, we control for external reference prices as competitors' prices with the variable $Competitorprice_{jt}$. This variable is defined as the percentage difference of the price of product j at day t from the median competition price for this product. In addition, we control for varying delivery times for different products as well as different product categories.

We estimate our reduced demand model using a Poisson regression and obtain a multiplicative model of the expected level of demand as a function of the previously mentioned covariates for each product and day. Given the Poisson distribution, this approach results in the following Poisson regression function:

$$Demand_{jt}|x_{jt} \sim f\left[\alpha_j * \exp\left(\boldsymbol{x}_{jt}^{'}\boldsymbol{\beta}\right)\right] = f\left[\exp\left(\ln(\alpha_j) + \boldsymbol{x}_{jt}^{'}\boldsymbol{\beta}\right)\right] \tag{11}$$

Here, a_j represents the unobserved product-specific random effects, which are supposed to be uncorrelated with the covariates $\boldsymbol{x}_{jt}$ and gamma-distributed. We estimate expected demand with the expected value of the observation-specific error term being zero:

$$E\left(Demand_{jt}|x_{jt}, \boldsymbol{\beta}\right) = \exp\left(\boldsymbol{X}_{\mathbf{i}}^{'}\boldsymbol{\beta}\right) = exp\begin{pmatrix} \beta_0 + \beta_1 Gain_{jt} + \beta_2 Loss_{jt} \\ + \beta_3 High.Amplitude_{jt} + \beta_4 High.Frequency_{jt} \\ + \beta_5 High.Amplitude_{jt} \times High.Frequency_{jt} \\ + \beta_6 High.Amplitude_{jt} \times Gain_{jt} \\ + \beta_7 High.Frequency_{jt} \times Gain_{jt} \\ + \beta_8 High.Amplitude_{jt} \times High.Frequency_{jt} \times Gain_{jt} \\ + \beta_9 High.Amplitude \times Loss_{jt} \\ + \beta_{10} High.Frequency \times Loss_{jt} \\ + \beta_{11} High.Amplitude_{jt} \times High.Frequency_{jt} \times Loss_{jt} \\ + \beta_{12} Competitorprice_{jt} + \boldsymbol{\beta_{13}} \boldsymbol{Deliverytime_{jt}} \\ + \boldsymbol{\beta_{14}} \boldsymbol{Categories_j} + ln(\alpha_j) + \varepsilon_{jt} \end{pmatrix} \tag{12}$$

[1] The median ratio of highest to lowest price of pre-experimental price variation is 1.155. This can be transferred to an amplitude of about 8%. The mean ratio of pre-experimental price changes per day and product is 0.308. This can be transferred to a frequency of about every third day.

Table 2
Descriptions and summary statistics of variables for the internal reference price and the reduced demand model.

Variables	Descriptions	N [b]	mean	median	sd	min	max
$Past.Prices_{jt}$	Weighted average of the last six prices for the same product as calculated in equation (9) (absolute in €)	110,573	773.55	287.22	1,202.61	2.90	15,529
$Price.Trend_j$	Average daily product-category price trend ($\frac{Price_t}{Price_{t-1}}$)	110,573	1.00	1.00	0.00	0.99	1.01
$Demand_{jt}$	Realized demand of product j on day t (absolute)	2,878	0.60	0	5.18	0	262
$High.Amplitude_{jt}$	1 for experimental factor high amplitude; 0 for experimental factor low amplitude of price changes	2,878	0.60	1	0.49	0	1
$High.Frequency_{jt}$	1 for experimental factor high frequency; 0 for experimental factor low frequency of price changes	2,878	0.50	0	0.50	0	1
$Ref.Price_{jt}$	Predicted internal reference price of product j at time t (absolute in €)	2,878	289.12	124.69	525.73	21.16	5,023.37
$Gain_{jt}$	$RefPrice_{jt} - Price_{jt}$ in the gain domain; 0 otherwise (absolute in €)	2,878	6.41	2.85	15.77	0	373.37
$Loss_{jt}$	$Price_{jt} - RefPrice_{jt}$ in the loss domain; 0 otherwise (absolute in €)	2,878	4.79	0	20.03	0	487.57
$Price_{jt}$	Price of product j at time t (absolute in €)	2,878	287.50	121.9	528.01	17.9	5,118
$Competitorprice_{jt}$	Percentage difference of $Price_{jt}$ to the median price of the competition for the same product j on the same day t	2,878	3.59	2.73	9.85	−44.58	84.90
$Fast.Delivery_{jt}$	1 for delivery time of product j at time t within 3 days; 0 otherwise [a]	2,878	0.89	1	0.31	0	1
$Medium.Delivery_{jt}$	1 for delivery time of product j at time t of 4–10 days; 0 otherwise [a]	2,878	0.02	0	0.15	0	1

Notes: [a] Baseline: $Slow.Delivery_{jt}$ for a delivery time of more than 10 days. [b] The internal reference price model estimation is based on 4,566 products, and the reduced demand model estimation is based on 181 products.

To calculate internal reference prices, we use the internal reference price model of equation (10) outlined above. We estimate internal reference prices with past prices and a product-category-specific price trend variable for 110 categories and 4,566 unique products. The price trend captures the average daily price changes across all products within one of 110 product categories. We exclude prices of the 181 products, which are manipulated in the field experiment, from the general category-specific price trend calculation. We capture the magnitude of price variation by $High.Frequency_{jt}$ and $High.Amplitude_{jt}$, both of which either equal one for the high experimental factor group or zero otherwise. Further, we calculate $Gain_{jt}$ and $Loss_{jt}$ for the internal reference price effects according to equations (1) and (2), as well as control for competitors' prices.

Table 2 describes all relevant variables and presents summary statistics.

4.3. Results

We first report the results of the internal reference price model and then the results of the reduced demand model. Table 3 depicts the results of the internal reference price model of equation (10). All coefficient estimates have the predicted sign. The coefficient of the weighted past prices is significant ($\alpha_1 = 0.995$ ($p < .01$)). The category price trend variable is low in variation and does not explain much variance of the internal reference price. However, it is conceptually meaningful, and for the prediction of the internal reference price, it is helpful to include it in the estimation, following Kalwani et al. (1990). Model fit is high with an overall $R^2 > .9$.

We estimate our reduced demand model of equation (11) with random effects and clustered standard errors at the product level. Table 4 depicts the results for the relevant variables and interaction terms. The demand estimation of equation (12) indicates that the general effect of the internal reference prices differs between the gain and loss domain. The effect is significant (and negative) in the loss domain ($\beta_2 = -0.0815$ ($p < .05$)) but insignificant in the gain domain ($\beta_1 = 0.0140$ ($p > .1$)). Since perceived losses (i.e., current price higher than internal reference price) have a significant negative effect on demand and outweigh perceived gains (i.e., current price lower than internal reference price), which have an insignificant effect, we find support for our expectation of asymmetric internal reference price effects as well as for proposition 1 that higher internal reference prices increase demand.

Table 3
Estimation of the internal reference price model of equation (10).

Variables	$Ref.Price_{jt}$
$Past.Prices_{jt}$	0.995***
	(0.001)
$Price.Trend_j$	−291.1
	(233.5)
Intercept	294.6
	(233.8)
Within R-squared	0.3961
Overall R-squared	0.9997
Observations	110,573
Number of products	4,566

Notes: Weight w = 0.4. Standard errors clustered on product level in parentheses. *** p < 0.01, ** p < 0.05, * p < 0.1.

Furthermore, we find that both *high amplitude* and *high frequency* attenuate the negative effect of internal reference prices in the loss domain. *High amplitude* attenuates the internal reference price effects in the loss domain with a coefficient of $\beta_9 = 0.0678$ ($p < .05$). Similarly, *high frequency* eradicates most of the internal reference price effect in the loss domain with a coefficient of $\beta_{10} = 0.0677$ ($p < .05$). Thus, *high amplitude* and *high frequency* both individually attenuate the internal reference price effects. However, there is no additional interaction of the two factors ($\beta_{11}, p > .05$).

The general internal reference price effect in the gain domain and all interactions with the two factors of dynamic pricing are insignificant. Nevertheless, the direction and relative size of the effects are consistent with the findings for the loss domain. Thus, our results are consistent with propositions 2 and 3 regarding the loss domain, i.e., that *high amplitude* and/or *high frequency* in price changes attenuates the impact of internal reference prices.

Further, the coefficient estimates of the control variables have the expected signs. The effect of the current price relative to *competitive prices* (i.e., the percentage difference of the current price to the median competitor price of the focal product) is negative. A product *delivery time* within three days increases demand relative to slower delivery times.

Table 4
Estimation of the reduced demand model of equation (12) for the field experiment.

Variables	Coefficient	Estimate
$Gain_{jt}$	β_1	0.0140
		(0.0181)
$Loss_{jt}$	β_2	−0.0815**
		(0.0326)
$High.Amplitude_{jt}$	β_3	−0.458
		(0.427)
$High.Frequency_{jt}$	β_4	−0.154
		(0.274)
$High.Amplitude_{jt} \times High.Frequency_{jt}$	β_5	0.279
		(0.538)
$High.Amplitude_{jt} \times Gain_{jt}$	β_6	−0.0123
		(0.0266)
$High.Frequency_{jt} \times Gain_{jt}$	β_7	−0.0147
		(0.0223)
$High.Amplitude_{jt} \times High.Frequency_{jt} \times Gain_{jt}$	β_8	0.0128
		(0.0269)
$High.Amplitude_{jt} \times Loss_{jt}$	β_9	0.0678**
		(0.0335)
$High.Frequency_{jt} \times Loss_{jt}$	β_{10}	0.0677**
		(0.0338)
$High.Amplitude_{jt} \times High.Frequency_{jt} \times Loss_{jt}$	β_{11}	−0.0677*
		(0.0372)
$Competitorprice_{jt}$	β_{12}	−0.0632*
		(0.0377)
$Fast.Delivery_{jt}$	β_{13}	1.404**
		(0.666)
$Medium.Delivery_{jt}$	β_{13}	0.899
		(0.627)
Intercept	β_0	−2.478***
		(0.388)
Observations		2,878
Number of products		181

Notes: Dependent variable: demand of product *j* on day *t*. Category controls included. Standard errors clustered at product level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5. Study 2: Online lab experiment

To test the propositions in a setting where we can measure participants' internal reference prices, additional control variables, and demographics, we conduct an online lab experiment in Study 2.

5.1. Experimental design and data

For the online lab experiment, we use a similar experimental setup as with the field experiment but use a between-subject design where treatments differ in price developments displayed to participants for the same product. The product is a generic laptop to allow for high comparability to the consumer electronics category of the field experiment. The experimental factors are, consistent with the field experiment, amplitude and frequency. Amplitude differentiates between high (~8%) and low (~5%) amplitude in price changes. Frequency is implemented with either high (every day) or low (every ~ 4th day) frequency in price changes. This results in a 2 (amplitude) × 2 (frequency) full-factorial between-subjects design consistent with the field experiment (see Fig. 1). Each of the four factor combinations consists of two variants a) and b) to increase variation in perceived gains and losses. In variant a) prices evolved such that 'yesterday's price' is lower and the average of the last six prices is $5 lower than today's price. Respectively, in variant b) 'yesterday's price' is higher and the average of the last six prices is $5 higher compared to today's price (see Table B1 in Appendix B for detailed price developments).

We collected a sample of 809 U.S.-based participants via the crowdsourcing platform prolific. Across all experimental groups, participants read a short scenario (see Fig. B1 in Appendix B for details): They were asked to imagine that they were looking for a new laptop, decided on the brand and model, and observed the prices of this laptop at a well-known online retailer for the last six days. Participants receive the prices of the last six days displayed sequentially on one page. This mimics the experience of price discovery-one day after another and allows at the same time to keep historic prices salient. Thereafter, participants observe "today's price" of $599 and need to state their purchase intention on a seven-point Likert scale (1 = "I would definitively not buy" to 7 = "I would definitely buy"). We measured participants' internal reference price with the open question "Based on the prices of the last six days at the online retailer I checked, I think the normal price for the laptop would be $___.", according to Chandrashekaran & Jagpal (1995).

In addition, we measured perceived retailer's quality focus with the question "The retailer of the laptop is likely to place emphasis on product quality", trust towards the retailer with the question "I believe that I could trust this retailer" (Darke, Brady, Benedicktus, & Wilson, 2016), and perceived retailer opportunism with the question "Regarding the retailer's pricing activities, I feel betrayed." (Martin, Borah, & Palmatier, 2017). Participants rated all three questions on a seven-point Likert scale (1 = "strongly disagree" to 7 = "strongly agree"). Finally, we measured demographics.

5.2. Results

We test our propositions by estimating an OLS model with robust standard errors and participants' purchase intentions as dependent variable including a similar set of predictor variables as in Study 2. Based on the measure of participants' internal reference price and the stated current price of $599, we calculated the *Gain* and *Loss* according to equations (1) and (2). Further, we include the direct effect of the indicator variables for the experimental groups, *high amplitude*, *high frequency*, their interaction, and their interactions with the continuous variables of gain and loss. We include controls for perceived *Quality*, *Trust*, *Opportunism*, as well as demographics (*Age,* and gender (1 for *Female*)).

The results displayed in Table 5 indicate that the effect of the internal reference prices differs between the gain and loss domain. The effect of *Gain* is insignificant while the effect of *Loss* is negative and significant ($\beta_2 = -0.00929$ ($p < .05$)). Since perceived losses (i.e., current price higher than internal reference price) have a significant negative effect on demand and outweigh perceived gains (i.e., current price lower than internal reference price), which have an insignificant effect, we again find support for our expectation of asymmetric reference price effects as well as for proposition 1.

Furthermore, we find that *high amplitude* attenuates the negative effect of internal reference prices in the loss domain. Since *high amplitude* weakens the internal reference price effect in the loss domain with a coefficient of $\beta_9 = -0.00877 (p = .055)$, we find partial support for proposition 2. We do not find *high frequency* to significantly attenuate the effect of internal reference prices in the loss domain ($\beta_{10} = -0.00840$ ($p > .1$)).

The internal reference price effect in the gain domain and all interactions with the two factors of dynamic pricing are insignificant. Thus, our results are consistent with proposition 2 regarding the loss domain. Nevertheless, we do not find additional support for proposition 3.

Table 5
Estimation of reduced demand model for the online lab experiment.

Variables	Coefficient	Estimate
Gain	β_1	−0.00101
		(0.00162)
Loss	β_2	−0.00929**
		(0.00409)
High.Amplitude	β_3	−0.0692
		(0.203)
High.Frequency	β_4	0.00981
		(0.231)
High.Amplitude × *High.Frequency*	β_5	−0.643*
		(0.335)
High.Amplitude × *Gain*	β_6	0.00214
		(0.00395)
High.Frequency × *Gain*	β_7	0.00619
		(0.0102)
High.Amplitude × *High.Frequency* × Gain	β_8	0.0105
		(0.0146)
High.Amplitude × *Loss*	β_9	0.00877*
		(0.00456)
High.Frequency × *Loss*	β_{10}	−0.00840
		(0.00854)
High.Amplitude × *High.Frequency* × Loss	β_{11}	0.00411
		(0.0101)
Quality	$\beta_{quality}$	0.130**
		(0.0551)
Trust	β_{trust}	0.281***
		(0.0532)
Opportunism	$\beta_{opportunism}$	0.00451
		(0.0386)
Age	β_{age}	0.00474
		(0.00760)
Female	β_{female}	−0.108
		(0.138)
Intercept	β_0	2.894***
		(0.446)
Observations		809
R^2		0.137

Notes: Dependent variable: Purchase intention on a Likert scale between 1 and 7. Robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Further, the covariates for *Quality* ($\beta_{quality} = 0.130$ ($p < .05$)) and *Trust* ($\beta_{trust} = 0.281$ ($p < .01$)) point in the expected direction such that participants who expect the retailer to place emphasis on product quality or participants who trust the retailer have higher purchase intentions.

6. Discussion

Technological innovations, such as algorithmic pricing, enable prices to change rapidly in nearly every product category and retailing context. Retailers benefit from dynamically adapting prices to current demand and supply conditions. In a competitive environment, dynamic pricing can extract additional consumer surplus, e.g., by selling to different customer segments over time (e.g., according to consumer price sensitivity). However, dynamic pricing can at the same time increase uncertainty in internal reference prices and therefore attenuate their impact on the decision of consumers (Haynes, 2009; Mazumdar & Jun, 1992).

To increase our understanding of this phenomenon, we derive propositions for the impact of internal reference prices on consumers' demand in a dynamic pricing environment from a reduced demand model. We tested the proposition in a field experiment at a durable products retailer and an online lab experiment. In the field experiment, we estimated the internal reference price based on past prices. In the online lab experiment, we measured the internal reference price of each consumer. For both experiments, we manipulated the amplitude and frequency of price changes.

6.1. Theoretical contribution

This study contributes to the literature on dynamic pricing and reference price effects as follows. In our experimental studies, we find asymmetric internal reference price effects on consumers' purchase behavior. Losses, i.e., current prices higher than the internal reference price, substantially reduce demand, while gains yield no significant effects. Therefore, our results for durable products are consistent with previous research that found asymmetric reference price effects for consumable products (e.g., Putler (1992)).

The field experiment shows that dynamic pricing attenuates the negative effect of losses in both dimensions, i.e., high amplitude and high frequency. The online lab experiment complements our analysis by observing participants' real internal reference prices. Results support the finding of the field experiment that high amplitude in prices attenuates the negative effect of losses.

Based on a theoretical model, Kopalle, Rao, & Assunção (1996) argue that if consumers weigh gains more than same-sized losses, price changes are optimal for the retailer. Otherwise, when consumers weigh losses more strongly than same-sized gains, a constant pricing policy is preferable. Consequently, our results support the expectation that consumers weigh losses more strongly than gains, leading to the expectation that price changes are disadvantageous for a retailer. However, dynamic pricing substantially reduces the unfavorable reference price effects. High amplitude and high frequency in price changes in the field experiment as well as high amplitude in the online lab experiment reduce the impact of internal reference prices, which can be explained by consumers being less certain about their internal reference price in a dynamic pricing environment (Mazumdar & Jun, 1992).

We can derive implications for price optimization based on a two-period model building on the demand model of Section 3 (see Appendix C). Based on this model, dynamic pricing alters the effect of unexpected cost shocks on optimal prices due to the attenuated reference price effects (compared to a situation without dynamic pricing): In case of a cost decreasing shock, optimal prices decrease to a lesser extent with dynamic pricing than without dynamic pricing. However, in case of a cost increasing shock, optimal prices increase to a larger extent with dynamic pricing than without dynamic pricing.

6.2. Managerial implications

Our findings have several managerial implications. First, our results underscore that retailers should be aware of internal reference price effects and that consumers weigh losses more strongly than gains. Second, our results indicate that dynamic pricing offers an additional benefit (i.e., in addition to the flexibility of adapting prices to current demand and supply conditions) by attenuating the negative internal reference price effects in the case of price increases. We find that both higher amplitude and higher frequency exert this attenuating effect in the field. Therefore, retailers may determine how to implement dynamic pricing via these two dimensions based on other considerations, e.g., related to supply chain management. Third, for contexts in which strong internal reference price effects might be preferable, retailers should decrease price variation. For example, in the case of a price skimming strategy (Spann et al., 2015), retailers might profit from higher internal reference prices than current prices (Crettez, Hayek, & Zaccour, 2020). In addition, conspicuous consumption products might profit from a high internal reference price that indicates the exclusivity of the product (Alford & Engelland, 2000). Therefore, high price variation appears less

preferable in the presence of a price skimming strategy or conspicuous consumption products.

Our results are not only informative for retailers. In addition, also consumers can benefit from being more aware of the effects of dynamically changing prices. Consumers should invest effort into price comparison not only between retailers but also across time. In addition to the assistance provided by price comparison websites, consumer advocacy groups could support consumers by providing accessible historic prices or by lobbying retailers to provide such prices.

6.3. Limitations and future research

We must acknowledge several limitations of our research that suggest avenues for future research. In the field experiment, we cannot examine whether consumers observed prices in the past and/or current competitor prices. We approached this challenge in two ways: First, we applied common assumptions for the formation of internal reference prices and chose the weights of past prices to maximize the explanatory power of our internal reference price model (equation (10)). Second, we replicated a similar experimental design in an online laboratory setting. Here, we were able to measure internal reference prices. Nevertheless, we must also acknowledge the specific limitations of the online lab experiment. The online lab experiment entailed non-consequential, hypothetical choices.

We study a durable products category, which is characterized by rather frequent new product introductions and rather short life cycles (Kelly et al., 2021; Spann et al., 2015), with internal reference prices rather based on search than past purchases. Future research may test our models in product categories with longer life cycles and with greater importance of past purchases for reference price formation.

Future research might further explore the impact of quality, trust, and opportunism towards the retailer in the presence of dynamic pricing. For example, previous literature has shown a strong correlation of price and quality perceptions for higher-priced products (Völckner & Hofmann, 2007) as well as a potentially negative effect of a low-price strategy on retailers' credibility (Hoch, Drèze, & Purk, 1994). Additionally, price fairness perceptions (Haws & Bearden, 2006) may be affected differently via amplitude and frequency. Further, future research may extend the two-period model used to illustrate the effect of dynamic pricing in the Appendix C to more general approaches such as multi-period models. Finally, future research may build on our systematization of dynamic prices according to amplitude and frequency in analyzing the effects of dynamic pricing. Additional studies could estimate the optimal amplitude and frequency of price changes by analyzing potential diminishing returns of dynamic prices on the reference price effects.

CRediT authorship contribution statement

David Prakash: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Martin Spann:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A1 shows that neither the non-experimental variables nor prices explain the allocation to the experimental factors, thus suggesting a clean randomization.

Table A1
Probit regressions for randomization test of the experimental factors of the field experiment.

Variables	$High.Amplitude_{jt}$	$High.Frequency_{jt}$
$Price_{jt}$	−0.000147	0.0000716
	(0.000550)	(0.000469)
$Fast.Delivery_{jt}$	−0.0232	−0.0256
	(0.904)	(1.049)
$Medium.Delivery_{jt}$	0.146	−0.0733
	(1.656)	(2.313)
$Product.Category.A_{jt}$	13.37	0.333
	(250.0)	(1.539)
$Product.Category.B_{jt}$	17.49	0.807
	(250.0)	(1.501)
$Product.Category.C_{jt}$	13.08	−2.386
	(250.0)	(5.898)
$Product.Category.D_{jt}$	13.33	0.351
	(250.0)	(1.527)
Constant	−13.01	−0.532
	(250.0)	(1.606)
Observations	2,878	2,878
Number of products	181	181

Notes: Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2 shows the internal reference price estimation of equation (10) for varying weights (from 0.1 to 0.9) in the calculation of $Past.Prices_{jt}$ for the field experiment. A weight of 0.4 yields the best fit.

Table A2
Internal reference price model estimations with $Ref.Price_{jt}$ as dependent variable and varying weights for the field experiment.

	weight								
Variables	0.1	0.2	0.3	**0.4**	0.5	0.6	0.7	0.8	0.9
$Past.Prices_{jt}$	0.994***	0.994***	0.995***	**0.995*** **	0.995***	0.996***	0.996***	0.996***	0.995***
	(0.00120)	(0.00113)	(0.00108)	**(0.00104)**	(0.00101)	(0.00101)	(0.00102)	(0.00104)	(0.00108)
$Price.Trend_j$	−862.9***	−684.1***	−496.7**	**−298.0**	−93.01	105.0	279.1	414.8	505.8
	(263.0)	(246.7)	(237.3)	**(235.0)**	(240.0)	(251.4)	(268.0)	(287.6)	(308.6)
Intercept	867.6***	688.4***	500.6**	**301.5**	96.21	−102.1	−276.3	−412.1	−503.0
	(263.6)	(247.3)	(237.7)	**(235.4)**	(240.2)	(251.7)	(268.2)	(287.8)	(308.9)
r-squared within	0.3779	0.3879	0.3942	**0.3960**	0.3926	0.3832	0.3676	0.3468	0.3227
r-squared overall	0.9997	0.9997	0.9997	**0.9997**	0.9997	0.9997	0.9997	0.9996	0.9996
Observations	110,573	110,573	110,573	**110,573**	110,573	110,573	110,573	110,573	110,573
No. of products	4,566	4,566	4,566	**4,566**	4,566	4,566	4,566	4,566	4,566

Notes: Standard errors clustered on product level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Appendix B

Table B1 shows the price developments in the online lab experiment by the factors 2 (frequency) × 2 (amplitude) in the variants a) and b). Price developments result in either (a) a negative or (b) a positive difference between the average and today's price as well as in the last step from yesterday's to today's price.

Table B1
Price development for each experimental group of the online lab experiment. Each group consists of two different price developments.

	High Frequency				**Low Frequency**			
	High Amplitude		**Low Amplitude**		**High Amplitude**		**Low Amplitude**	
	(a)	(b)	(a)	(b)	(a)	(b)	(a)	(b)
six days ago	639 €	539 €	619 €	559 €	639 €	559 €	619 €	579 €
five days ago	549 €	649 €	589 €	609 €	639 €	559 €	619 €	579 €
four days ago	659 €	559 €	629 €	569 €	639 €	559 €	619 €	579 €
three days ago	559 €	639 €	579 €	619 €	549 €	649 €	569 €	629 €
two days ago	649 €	549 €	639 €	579 €	549 €	649 €	569 €	629 €
yesterday	569 €	629 €	569 €	629 €	549 €	649 €	569 €	629 €
today	599 €	599 €	599 €	599 €	599 €	599 €	599 €	599 €

Fig. B1 depicts the experimental stimuli and dependent variable question for a participant of group one. Participants received an identical text with different prices, depending on their randomly allocated group, as displayed in Table B1.

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	**639 €**

continue

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	639 €
five days ago	**549 €**

continue

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	639 €
five days ago	549 €
four days ago	**659 €**

continue

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	639 €
five days ago	549 €
four days ago	659 €
three days ago	**559 €**

continue

Fig. B1. Experimental Stimuli (Online Lab Experiment).

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	639 €
five days ago	549 €
four days ago	659 €
three days ago	559 €
two days ago	**649 €**

continue

Imagine that you have been wanting **a new laptop** and have decided exactly what brand and model you will buy. For **the last six days** you **checked the price** for the laptop of a **well-known online retailer.**

The price developed as following:

Checked prices for the laptop	
six days ago	639 €
five days ago	549 €
four days ago	659 €
three days ago	559 €
two days ago	649 €
yesterday	**569 €**

Continue to the questions… **continue**

Today, you return to the well-known online retailer and **the laptop costs $599.**

Given that you want a new laptop, **how likely are you to buy** this laptop for **today's price of $599?**

I would definitely not buy	o	o	o	o	o	o	o	I would definitely buy

continue

Fig. B1. (*continued*).

Appendix C

We develop a two-period model based on the demand function (5) accounting for the possibility of an unexpected cost shock in period 2. The model assumes that prior use of dynamic pricing mitigates the reference price effect. The model shows that dynamic pricing alters the impact of cost shocks on optimal prices: In the case of a cost decreasing shock, optimal prices decrease to a lesser extent with dynamic pricing than without dynamic pricing. However, in case of a cost increasing shock, optimal prices increase to a larger extent with dynamic pricing than without dynamic pricing.

Setting: In period 1, consumers have no internal reference price, e.g., because the product is new to the market. In period 2, consumers have an internal reference price equal to first period's price ($RP_2 = p_1$). Thus, in addition to the base demand a, the optimal price of period 2 depends on the

price of period 1 and the cost c_2. These costs may differ from period 1 in case of an unexpected cost shock.

Period 2: Profit accounts for possible loss-aversion $\lambda \geq 1$ and prior dynamic pricing captured via amplitude $\sigma_j^a \in [0,1]$ and frequency $\sigma_j^f \in [0,1]$. For ease of exposition, we denote prior dynamic pricing in the following with $N = \left(1-\sigma_j^a\right)*\left(1-\sigma_j^f\right)$, i.e., *no* prior dynamic pricing ($\sigma_j^a = 0 \wedge \sigma_j^f = 0$) is denoted as N = 1:

$$\pi_2 = (p_2 - c_2)*(a - p_2 + \lambda*N*(RP_2 - p_2)) \text{ with } RP_2 = p_1$$

$$\frac{\partial \pi_2}{\partial p_2} = 0 \rightarrow p_2 = \frac{1}{2}\left(\frac{a + p_1*\lambda N}{1+\lambda N} + c_2\right)$$

Period 1: The price in the first period is determined by taking the optimal price of the second period into account when calculating the total profit π_T including a discount factor $\delta \in [0,1]$ for second period's profit:

$$\pi_T = (p_1 - c_1)*(a - p_1) + \delta*(p_2 - c_2)*(a - p_2 + \lambda N*(p_1 - p_2))$$

$$\pi_T = (p_1 - c_1)*(a - p_1) + \delta*\left(\frac{1}{2}\left(\frac{a + p_1*\lambda N}{1+\lambda N} + c_2\right) - c_2\right)*\left(a - \frac{1}{2}\left(\frac{a + p_1*\lambda N}{1+\lambda N} + c_2\right) + \lambda N\left(p_1 - \frac{1}{2}\left(\frac{a + p_1*\lambda N}{1+\lambda N} + c_2\right)\right)\right)$$

$$\frac{\partial \pi_T}{\partial p_1} = 0 \rightarrow p_1 = \frac{2a + 2c_1 + 2\lambda Na + 2\lambda Nc_1 + \lambda N\delta a - \lambda N\delta c_2 - \lambda^2 N^2 \delta c_2}{4\lambda N + 4 - \delta\lambda^2 N^2}$$

For the case of no expected cost shock ($c_1 = c_2$), we can replace c_2 with c_1:

$$p^*_{1(c_1=c_2)} = \frac{2a + 2c_1 + 2\lambda Na + 2\lambda Nc_1 + \lambda N\delta a - \lambda Ndc_1 - \lambda^2 N^2 dc_1}{4\lambda N + 4 - d\lambda^2 N^2}$$

Substituting $p^*_{1(c_1=c_2)}$ for p_1 in p_2 leads to the optimal price in period 2:

$$p^*_2(p^*_{1(c_1=c_2)}) = \frac{4a + 4c_2 + 2\lambda Na + 2\lambda Nc_1 + 4\lambda Nc_2 - \lambda^2 N^2 \delta c_2 - \lambda^2 N^2 \delta c_1}{8\lambda N + 8 - 2\delta\lambda^2 N^2}$$

Impact of unexpected cost shocks on optimal prices

We analyze the effect of a cost decreasing and increasing cost shock on the optimal prices for the extreme cases of high dynamic pricing, i.e., N = 0: ($\sigma_j^f \rightarrow 1$ *and/or* $\sigma_j^f \rightarrow 1$) and no dynamic pricing, i.e., N = 1: ($\sigma_j^a = 0 \wedge \sigma_j^f = 0$).

1) Cost decreasing shock ($c_1 > c_2$)

With and without dynamic pricing, the price in the second period is lower than in period one for a cost decreasing shock, hence loss aversion λ does not play a role:

$$p^*_{1(c_1=c_2)} > p^*_{2(c_1>c_2)}\left(p^*_{1(c_1=c_2)};\ \sigma_j^a = 1 \text{ and}/\text{or } \sigma_j^f = 1\right)$$

$$\frac{2a + 2c_1 + 2\lambda Na + 2\lambda Nc_1 + \lambda N\delta a - \lambda N\delta c_1 - \lambda^2 N^2 \delta c_1}{4\lambda N + 4 - \delta\lambda^2 N^2} > \frac{a + c_2}{2}$$

$$(4c_1 - 4c_2) + (4\lambda Nc_1 - 4\lambda Nc_2) + (2\lambda N\delta a - 2\lambda N\delta c_1) + (\lambda^2 N^2 \delta a + \lambda^2 N^2 \delta c_2 - 2\lambda^2 N^2 \delta c_1) > 0$$

For no dynamic pricing, i.e., N = 1: $p^*_{2(c_1>c_2)}\left(p^*_{1(c_1=c_2)}\right) = \frac{6a+8c_2+2c_1-\delta c_1-\delta c_2}{16-2\delta}$

For high dynamic pricing, i.e., N = 0: $p^*_{2(c_1>c_2)}\left(p^*_{1(c_1=c_2)}\right) = \frac{a+c_2}{2}$

Due to the absence of a positive reference price effect for high dynamic pricing, the price decreases less for a cost decreasing shock with high dynamic pricing:

$$p^*_{2(c_1>c_2)}\left(p^*_{1(c_1=c_2)};\ \sigma_j^a = 0 \text{ and } \sigma_j^f = 0\right) \left\langle p^*_{2(c_1>c_2)}\left(p^*_{1(c_1=c_2)};\ \sigma_j^a = 1 \text{ and}/\text{or } \sigma_j^f = 1\right)\right.$$

$$\frac{6a + 8c_2 + 2c_1 - \delta c_1 - \delta c_2}{16 - 2\delta} < \frac{a + c_2}{2}$$

$$(2 - \delta)(c_1 - a) < 0$$

This is true since profit is non-negative leading to $c_1 < a$.

2) Cost increasing shock $c_1 < c_2$

Due to the absence of a negative reference price effect for high dynamic pricing, the price increases stronger for a cost increasing shock with high dynamic pricing:

$$p^*_{2(c_1<c_2)}\left(p^*_{1(c_1=c_2)};\ \sigma_j^a = 0 \text{ and } \sigma_j^f = 0\right) \left\langle p^*_{2(c_1<c_2)}\left(p^*_{1(c_1=c_2)};\ \sigma_j^a = 1 \text{ and}/\text{or } \sigma_j^f = 1\right)\right.$$

$$\left(\frac{4a+4c_2+2\lambda Na+2\lambda Nc_1+4\lambda Nc_2-\lambda^2N^2\delta c_2-\lambda^2N^2\delta c_1}{8\lambda N+8-2\delta\lambda^2N^2}\right)\left\langle\frac{a+c_2}{2}\right.$$

$$(2-\lambda N\delta)(c_1-a)<0$$

This is true since profit is non-negative leading to $c_1 < a$ and for not too high loss aversion $\lambda N\delta < 2$.

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